

NYPD Shooting Incident Data Report

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Load Required Packages

```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.6
```

```
## v forcats    1.0.1      v stringr    1.6.0
```

```
## v ggplot2    4.0.1      v tibble     3.3.0
```

```
## v lubridate  1.9.4      v tidyr      1.3.2
```

```
## v purrr      1.2.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Data Import

Data obtained from <https://data.cityofnewyork.us/api/views/833y-fsy8/rows.csv?accessType=DOWNLOAD>

```
nypd_csv_path <- "https://data.cityofnewyork.us/api/views/833y-fsy8/rows.csv?accessType=DOWNLOAD"
```

```
nypd_data <- read_csv(nypd_csv_path, skip_empty_rows = TRUE)
```

```
## Rows: 29744 Columns: 21
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr  (12): OCCUR_DATE, BORO, LOC_OF_OCCUR_DESC, LOC_CLASSFCTN_DESC, LOCATION...
```

```
## dbl  (5): INCIDENT_KEY, PRECINCT, JURISDICTION_CODE, Latitude, Longitude
```

```
## num  (2): X_COORD_CD, Y_COORD_CD
```

```
## lgl  (1): STATISTICAL_MURDER_FLAG
```

```
## time (1): OCCUR_TIME
```

```
##
```

```
## i Use 'spec()' to retrieve the full column specification for this data.
```

```
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(nypd_data)
```

```
## # A tibble: 6 x 21
##   INCIDENT_KEY OCCUR_DATE OCCUR_TIME BORO      LOC_OF_OCCUR_DESC PRECINCT
##   <dbl> <chr>      <time>    <chr>    <chr>          <dbl>
## 1   231974218 08/09/2021 01:06    BRONX    <NA>           40
## 2   177934247 04/07/2018 19:48    BROOKLYN <NA>           79
## 3   255028563 12/02/2022 22:57    BRONX    OUTSIDE        47
## 4   25384540 11/19/2006 01:50    BROOKLYN <NA>           66
## 5    72616285 05/09/2010 01:58    BRONX    <NA>           46
## 6    85875439 07/22/2012 21:35    BRONX    <NA>           42
## # i 15 more variables: JURISDICTION_CODE <dbl>, LOC_CLASSFCTN_DESC <chr>,
## #   LOCATION_DESC <chr>, STATISTICAL_MURDER_FLAG <lgl>, PERP_AGE_GROUP <chr>,
## #   PERP_SEX <chr>, PERP_RACE <chr>, VIC_AGE_GROUP <chr>, VIC_SEX <chr>,
## #   VIC_RACE <chr>, X_COORD_CD <dbl>, Y_COORD_CD <dbl>, Latitude <dbl>,
## #   Longitude <dbl>, Lon_Lat <chr>
```

```
summary(nypd_data)
```

```
##   INCIDENT_KEY      OCCUR_DATE      OCCUR_TIME
##   Min.   : 9953245   Length:29744   Min.   :00:00:00.000000
##   1st Qu.: 67321140   Class :character 1st Qu.:03:30:45.000000
##   Median :109291972   Mode  :character  Median :15:15:00.000000
##   Mean   :133850951                Mean   :12:46:10.874798
##   3rd Qu.:214741917                3rd Qu.:20:44:00.000000
##   Max.   :299462478                Max.   :23:59:00.000000
##
##   BORO      LOC_OF_OCCUR_DESC      PRECINCT      JURISDICTION_CODE
##   Length:29744   Length:29744   Min.   : 1.00   Min.   :0.0000
##   Class :character  Class :character 1st Qu.: 44.00   1st Qu.:0.0000
##   Mode  :character  Mode  :character Median : 67.00   Median :0.0000
##                                     Mean  : 65.23   Mean  :0.3181
##                                     3rd Qu.: 81.00   3rd Qu.:0.0000
##                                     Max.   :123.00   Max.   :2.0000
##                                     NA's   :2
##   LOC_CLASSFCTN_DESC LOCATION_DESC      STATISTICAL_MURDER_FLAG
##   Length:29744   Length:29744   Mode :logical
##   Class :character  Class :character FALSE:23979
##   Mode  :character  Mode  :character TRUE :5765
##
##
##
##   PERP_AGE_GROUP      PERP_SEX      PERP_RACE      VIC_AGE_GROUP
##   Length:29744   Length:29744   Length:29744   Length:29744
##   Class :character  Class :character  Class :character  Class :character
##   Mode  :character  Mode  :character  Mode  :character  Mode  :character
##
##
##
##   VIC_SEX      VIC_RACE      X_COORD_CD      Y_COORD_CD
```

```
## Length:29744      Length:29744      Min.   : 914928      Min.   :125757
## Class :character  Class :character  1st Qu.:1000094      1st Qu.:183042
## Mode  :character  Mode  :character  Median :1007826      Median :195506
##                                     Mean  :1009442      Mean  :208722
##                                     3rd Qu.:1016739      3rd Qu.:239980
##                                     Max.   :1066815      Max.   :271128
##
##      Latitude      Longitude      Lon_Lat
## Min.   :40.51      Min.   : -74.25      Length:29744
## 1st Qu.:40.67      1st Qu.: -73.94      Class :character
## Median :40.70      Median : -73.91      Mode  :character
## Mean   :40.74      Mean   : -73.91
## 3rd Qu.:40.83      3rd Qu.: -73.88
## Max.   :40.91      Max.   : -73.70
## NA's   :97         NA's   :97
```

```
unique(nypd_data$VIC_RACE)
```

```
## [1] "BLACK"                                "WHITE HISPANIC"
## [3] "BLACK HISPANIC"                      "ASIAN / PACIFIC ISLANDER"
## [5] "WHITE"                                "UNKNOWN"
## [7] "AMERICAN INDIAN/ALASKAN NATIVE"
```

Clean Data

```
# Get the number of PERP on VIC crime
nypd_data_clean <- nypd_data %>%
  filter(!is.na(PERP_RACE)) %>%
  filter(!is.na(VIC_RACE)) %>%
  filter(!(PERP_RACE %in% c('(null)', 'UNKNOWN')) %>%
  filter(!(VIC_RACE %in% c('(null)', 'UNKNOWN')) %>%
  mutate(PERP_ON_VIC_RACES = paste0(PERP_RACE, '>', VIC_RACE))

nypd_data_clean
```

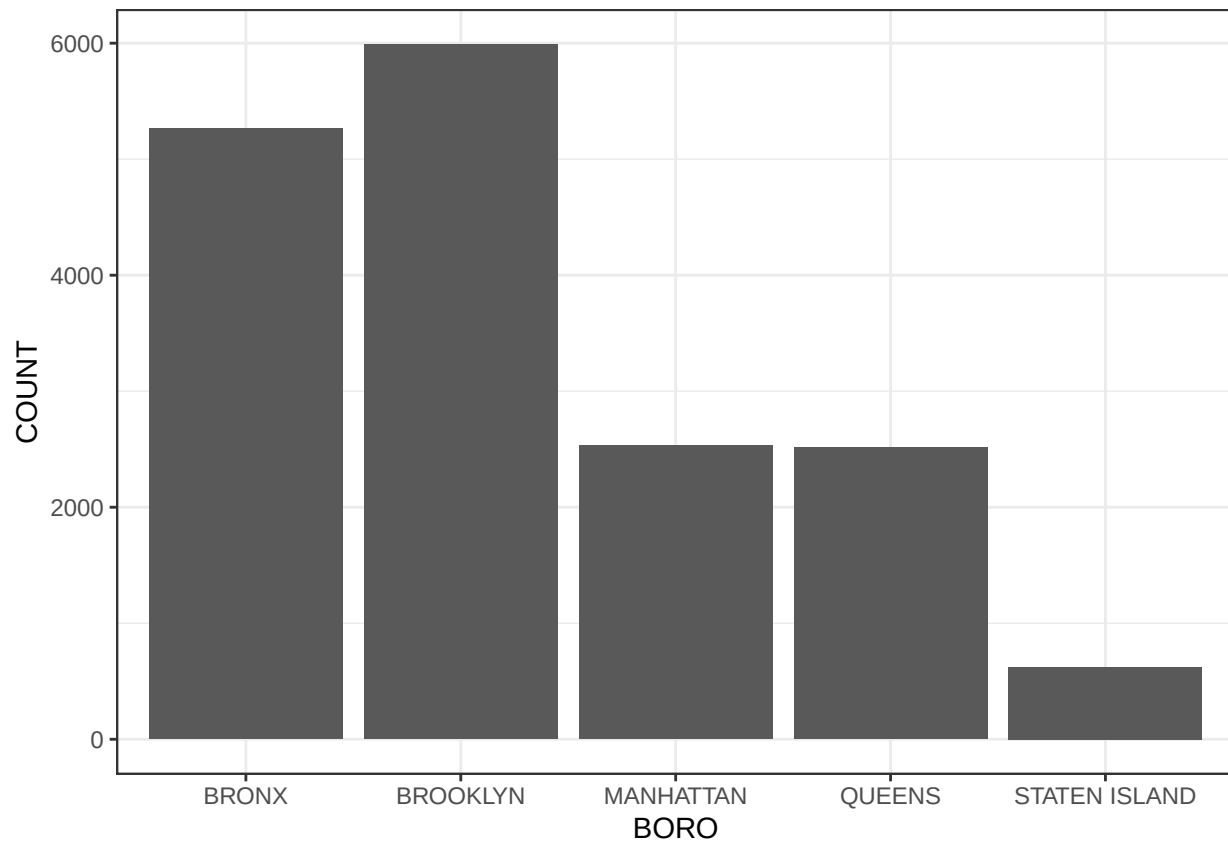
```
## # A tibble: 16,922 x 22
##   INCIDENT_KEY OCCUR_DATE OCCUR_TIME BORO      LOC_OF_OCCUR_DESC PRECINCT
##   <dbl> <chr>      <time> <chr>      <chr>      <dbl>
## 1  177934247 04/07/2018 19:48  BROOKLYN  <NA>      79
## 2   72616285 05/09/2010 01:58  BRONX     <NA>      46
## 3   85875439 07/22/2012 21:35  BRONX     <NA>      42
## 4  142324890 04/21/2015 15:36  BROOKLYN  <NA>      75
## 5  152868707 05/07/2016 15:23  BROOKLYN  <NA>      69
## 6   25036119 10/30/2006 00:20  BRONX     <NA>      40
## 7   69758511 01/10/2010 02:08  BRONX     <NA>      42
## 8   61459877 05/06/2009 22:20  BROOKLYN  <NA>      71
## 9   79489506 06/24/2011 04:36  BRONX     <NA>      47
## 10  38936881 02/05/2008 20:30  MANHATTAN <NA>      34
## # i 16,912 more rows
## # i 16 more variables: JURISDICTION_CODE <dbl>, LOC_CLASSFCTN_DESC <chr>,
## #   LOCATION_DESC <chr>, STATISTICAL_MURDER_FLAG <lgl>, PERP_AGE_GROUP <chr>,
```

```
## #   PERP_SEX <chr>, PERP_RACE <chr>, VIC_AGE_GROUP <chr>, VIC_SEX <chr>,  
## #   VIC_RACE <chr>, X_COORD_CD <dbl>, Y_COORD_CD <dbl>, Latitude <dbl>,  
## #   Longitude <dbl>, Lon_Lat <chr>, PERP_ON_VIC_RACES <chr>
```

Data Analysis

Murders by Borough

```
murder_by_borough <- nypd_data_clean %>%  
  group_by(BORO) %>%  
  count(BORO, name = 'COUNT')  
  
murder_by_borough %>%  
  ggplot(aes(x = BORO, y = COUNT)) +  
  geom_bar(stat = 'identity') +  
  theme_bw()
```

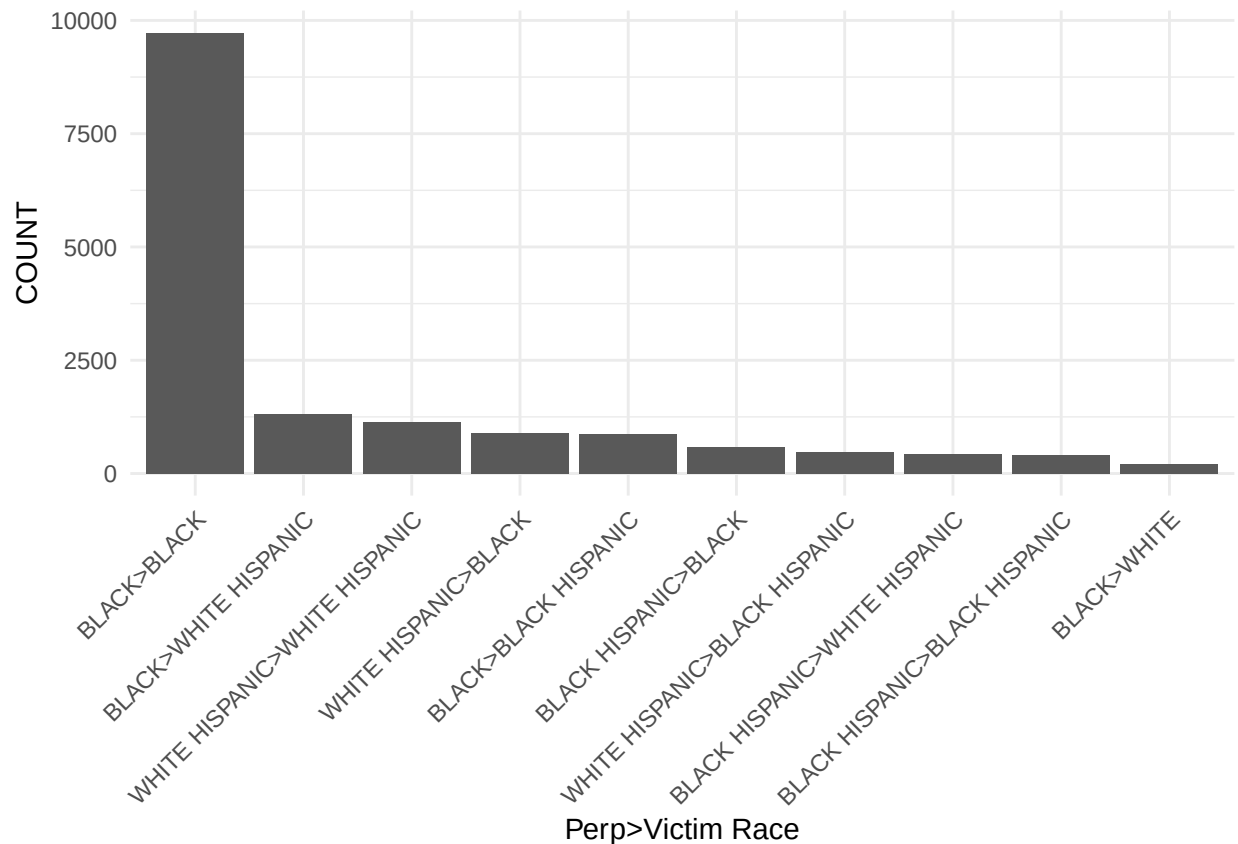


Race on Race Crime

```
race_on_race <- nypd_data_clean %>%  
  count(PERP_ON_VIC_RACES, name = 'COUNT') %>%  
  arrange(desc(COUNT)) %>%
```

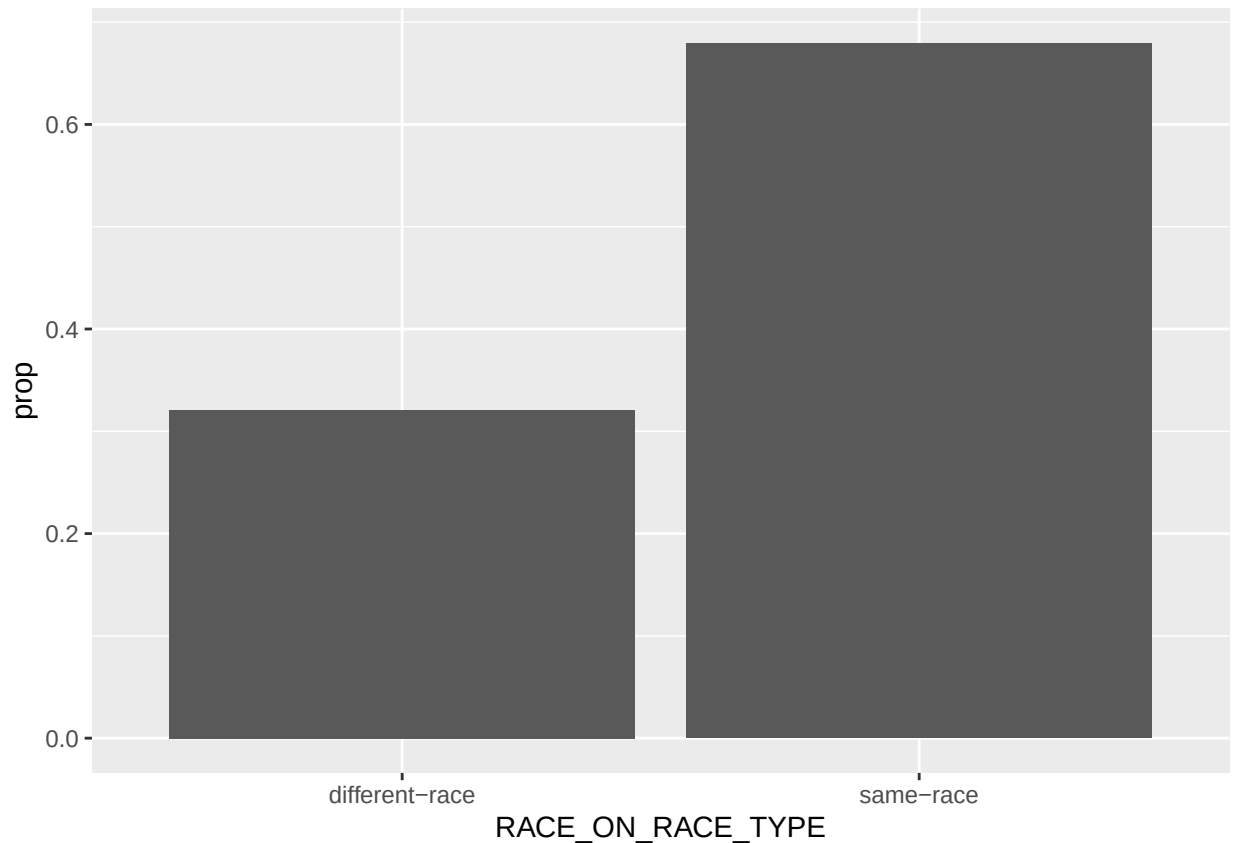
```
head(10)

race_on_race %>%
  ggplot(aes(x = reorder(PERP_ON_VIC_RACES, -COUNT), y = COUNT)) +
  geom_bar(stat = 'identity') +
  theme_minimal() +
  theme(axis.text.x = element_text(angle=45, hjust=1)) +
  xlab('Perp>Victim Race')
```



What is the proportion on same-race versus difference-race race-on-race crime?

```
nypd_data_clean %>%
  count(PERP_ON_VIC_RACES, name = 'COUNT') %>%
  arrange(desc(COUNT)) %>%
  mutate(PERP_RACE = gsub(">.*", "", PERP_ON_VIC_RACES)) %>%
  mutate(VICTIM_RACE = gsub(".*>", "", PERP_ON_VIC_RACES)) %>%
  mutate(RACE_ON_RACE_TYPE = ifelse(PERP_RACE == VICTIM_RACE, "same-race", "different-race")) %>%
  group_by(RACE_ON_RACE_TYPE) %>%
  summarize(count = sum(COUNT)) %>%
  mutate(prop = count / sum(count)) %>%
  ggplot(aes(x = RACE_ON_RACE_TYPE, y = prop)) +
  geom_bar(stat='identity')
```



Data Modelling

Is the increased prevalence of same-race crime consistent across all boroughs?

```
df <- nypd_data_clean %>%
  count(PERP_ON_VIC_RACES, BORO, name = "COUNT") %>%
  mutate(
    PERP_RACE = sub(">.*", "", PERP_ON_VIC_RACES),
    VICTIM_RACE = sub(".*>", "", PERP_ON_VIC_RACES),
    RACE_ON_RACE_TYPE = ifelse(PERP_RACE == VICTIM_RACE,
                              "same-race", "different-race")
  )
head(df)
```

```
## # A tibble: 6 x 6
##   PERP_ON_VIC_RACES      BORO  COUNT PERP_RACE VICTIM_RACE RACE_ON_RACE_TYPE
##   <chr>                <chr> <int> <chr>      <chr>      <chr>
## 1 AMERICAN INDIAN/ALASKAN N~ BRONX     1 AMERICAN~ BLACK      different-race
## 2 AMERICAN INDIAN/ALASKAN N~ QUEE~     1 AMERICAN~ BLACK      different-race
## 3 ASIAN / PACIFIC ISLANDER>~ BRONX     3 ASIAN / ~ ASIAN / PA~ same-race
## 4 ASIAN / PACIFIC ISLANDER>~ BROO~    15 ASIAN / ~ ASIAN / PA~ same-race
## 5 ASIAN / PACIFIC ISLANDER>~ MANH~     5 ASIAN / ~ ASIAN / PA~ same-race
## 6 ASIAN / PACIFIC ISLANDER>~ QUEE~    45 ASIAN / ~ ASIAN / PA~ same-race
```

```

model <- glm(
  COUNT ~ BORO + PERP_RACE + VICTIM_RACE + PERP_RACE:VICTIM_RACE,
  family = poisson,
  data = df
)

summary(model)

```

```

##
## Call:
## glm(formula = COUNT ~ BORO + PERP_RACE + VICTIM_RACE + PERP_RACE:VICTIM_RACE,
##      family = poisson, data = df)
##
## Coefficients: (8 not defined because of singularities)
##
## (Intercept) -4.5932035
## BOROBROOKLYN 0.1284380
## BOROMANHATTAN -0.7324770
## BOROQUEENS -0.7409088
## BOROSTATEN ISLAND -2.1338531
## PERP_RACEASIAN / PACIFIC ISLANDER 1.6756575
## PERP_RACEBLACK 5.8128448
## PERP_RACEBLACK HISPANIC 4.3766486
## PERP_RACEWHITE 2.3132441
## PERP_RACEWHITE HISPANIC 5.3341123
## VICTIM_RACEASIAN / PACIFIC ISLANDER 1.9420871
## VICTIM_RACEBLACK 4.8965540
## VICTIM_RACEBLACK HISPANIC 4.2489185
## VICTIM_RACEWHITE 2.7458999
## VICTIM_RACEWHITE HISPANIC 5.1201409
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEASIAN / PACIFIC ISLANDER 4.0424137
## PERP_RACEBLACK:VICTIM_RACEASIAN / PACIFIC ISLANDER 0.8301753
## PERP_RACEBLACK HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER 0.3229120
## PERP_RACEWHITE:VICTIM_RACEASIAN / PACIFIC ISLANDER 2.1085501
## PERP_RACEWHITE HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK 0.9647142
## PERP_RACEBLACK:VICTIM_RACEBLACK 1.8997250
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK 0.5278740
## PERP_RACEWHITE:VICTIM_RACEBLACK 0.0229162
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK HISPANIC 0.1780753
## PERP_RACEBLACK:VICTIM_RACEBLACK HISPANIC 0.1247033
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK HISPANIC 0.7894461
## PERP_RACEWHITE:VICTIM_RACEBLACK HISPANIC -0.0006165
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK HISPANIC NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE 1.4894011
## PERP_RACEBLACK:VICTIM_RACEWHITE 0.2191653
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE -0.0855787
## PERP_RACEWHITE:VICTIM_RACEWHITE 3.4908719
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE HISPANIC NA
## PERP_RACEBLACK:VICTIM_RACEWHITE HISPANIC -0.3268640
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE HISPANIC NA

```

## PERP_RACEWHITE:VICTIM_RACEWHITE HISPANIC	NA
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE HISPANIC	NA
##	Std. Error
## (Intercept)	1.2254656
## BOROBROOKLYN	0.0188966
## BOROMANHATTAN	0.0241879
## BOROQUEENS	0.0242471
## BOROSTATEN ISLAND	0.0424342
## PERP_RACEASIAN / PACIFIC ISLANDER	0.7333326
## PERP_RACEBLACK	1.3043955
## PERP_RACEBLACK HISPANIC	0.7101930
## PERP_RACEWHITE	0.7212812
## PERP_RACEWHITE HISPANIC	0.7079394
## VICTIM_RACEASIAN / PACIFIC ISLANDER	1.0107500
## VICTIM_RACEBLACK	1.0007236
## VICTIM_RACEBLACK HISPANIC	1.0012281
## VICTIM_RACEWHITE	1.0049193
## VICTIM_RACEWHITE HISPANIC	1.0006126
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEASIAN / PACIFIC ISLANDER	0.2690164
## PERP_RACEBLACK:VICTIM_RACEASIAN / PACIFIC ISLANDER	1.1078327
## PERP_RACEBLACK HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER	0.2571799
## PERP_RACEWHITE:VICTIM_RACEASIAN / PACIFIC ISLANDER	0.3424832
## PERP_RACEWHITE HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER	NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK	0.2326052
## PERP_RACEBLACK:VICTIM_RACEBLACK	1.0961209
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK	0.0775127
## PERP_RACEWHITE:VICTIM_RACEBLACK	0.2059107
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK	NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK HISPANIC	0.3318856
## PERP_RACEBLACK:VICTIM_RACEBLACK HISPANIC	1.0970629
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK HISPANIC	0.0884359
## PERP_RACEWHITE:VICTIM_RACEBLACK HISPANIC	0.2542936
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK HISPANIC	NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE	0.3598031
## PERP_RACEBLACK:VICTIM_RACEWHITE	1.1020584
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE	0.1993661
## PERP_RACEWHITE:VICTIM_RACEWHITE	0.1858616
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE	NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE HISPANIC	NA
## PERP_RACEBLACK:VICTIM_RACEWHITE HISPANIC	1.0963201
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE HISPANIC	NA
## PERP_RACEWHITE:VICTIM_RACEWHITE HISPANIC	NA
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE HISPANIC	NA
##	z value
## (Intercept)	-3.748
## BOROBROOKLYN	6.797
## BOROMANHATTAN	-30.283
## BOROQUEENS	-30.557
## BOROSTATEN ISLAND	-50.286
## PERP_RACEASIAN / PACIFIC ISLANDER	2.285
## PERP_RACEBLACK	4.456
## PERP_RACEBLACK HISPANIC	6.163
## PERP_RACEWHITE	3.207
## PERP_RACEWHITE HISPANIC	7.535

## VICTIM_RACEASIAN / PACIFIC ISLANDER	1.921
## VICTIM_RACEBLACK	4.893
## VICTIM_RACEBLACK HISPANIC	4.244
## VICTIM_RACEWHITE	2.732
## VICTIM_RACEWHITE HISPANIC	5.117
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEASIAN / PACIFIC ISLANDER	15.027
## PERP_RACEBLACK:VICTIM_RACEASIAN / PACIFIC ISLANDER	0.749
## PERP_RACEBLACK HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER	1.256
## PERP_RACEWHITE:VICTIM_RACEASIAN / PACIFIC ISLANDER	6.157
## PERP_RACEWHITE HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER	NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK	4.147
## PERP_RACEBLACK:VICTIM_RACEBLACK	1.733
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK	6.810
## PERP_RACEWHITE:VICTIM_RACEBLACK	0.111
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK	NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK HISPANIC	0.537
## PERP_RACEBLACK:VICTIM_RACEBLACK HISPANIC	0.114
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK HISPANIC	8.927
## PERP_RACEWHITE:VICTIM_RACEBLACK HISPANIC	-0.002
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK HISPANIC	NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE	4.139
## PERP_RACEBLACK:VICTIM_RACEWHITE	0.199
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE	-0.429
## PERP_RACEWHITE:VICTIM_RACEWHITE	18.782
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE	NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE HISPANIC	NA
## PERP_RACEBLACK:VICTIM_RACEWHITE HISPANIC	-0.298
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE HISPANIC	NA
## PERP_RACEWHITE:VICTIM_RACEWHITE HISPANIC	NA
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE HISPANIC	NA
##	Pr(> z)
## (Intercept)	0.000178
## BOROBROOKLYN	1.07e-11
## BOROMANHATTAN	< 2e-16
## BOROQUEENS	< 2e-16
## BOROSTATEN ISLAND	< 2e-16
## PERP_RACEASIAN / PACIFIC ISLANDER	0.022313
## PERP_RACEBLACK	8.34e-06
## PERP_RACEBLACK HISPANIC	7.16e-10
## PERP_RACEWHITE	0.001341
## PERP_RACEWHITE HISPANIC	4.89e-14
## VICTIM_RACEASIAN / PACIFIC ISLANDER	0.054677
## VICTIM_RACEBLACK	9.93e-07
## VICTIM_RACEBLACK HISPANIC	2.20e-05
## VICTIM_RACEWHITE	0.006286
## VICTIM_RACEWHITE HISPANIC	3.10e-07
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEASIAN / PACIFIC ISLANDER	< 2e-16
## PERP_RACEBLACK:VICTIM_RACEASIAN / PACIFIC ISLANDER	0.453635
## PERP_RACEBLACK HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER	0.209265
## PERP_RACEWHITE:VICTIM_RACEASIAN / PACIFIC ISLANDER	7.43e-10
## PERP_RACEWHITE HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER	NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK	3.36e-05
## PERP_RACEBLACK:VICTIM_RACEBLACK	0.083072
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK	9.75e-12

## PERP_RACEWHITE:VICTIM_RACEBLACK	0.911385
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK	NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK HISPANIC	0.591574
## PERP_RACEBLACK:VICTIM_RACEBLACK HISPANIC	0.909499
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK HISPANIC	< 2e-16
## PERP_RACEWHITE:VICTIM_RACEBLACK HISPANIC	0.998066
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK HISPANIC	NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE	3.48e-05
## PERP_RACEBLACK:VICTIM_RACEWHITE	0.842365
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE	0.667738
## PERP_RACEWHITE:VICTIM_RACEWHITE	< 2e-16
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE	NA
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE HISPANIC	NA
## PERP_RACEBLACK:VICTIM_RACEWHITE HISPANIC	0.765591
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE HISPANIC	NA
## PERP_RACEWHITE:VICTIM_RACEWHITE HISPANIC	NA
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE HISPANIC	NA
##	
## (Intercept)	***
## BOROBROOKLYN	***
## BOROMANHATTAN	***
## BOROQUEENS	***
## BOROSTATEN ISLAND	***
## PERP_RACEASIAN / PACIFIC ISLANDER	*
## PERP_RACEBLACK	***
## PERP_RACEBLACK HISPANIC	***
## PERP_RACEWHITE	**
## PERP_RACEWHITE HISPANIC	***
## VICTIM_RACEASIAN / PACIFIC ISLANDER	.
## VICTIM_RACEBLACK	***
## VICTIM_RACEBLACK HISPANIC	***
## VICTIM_RACEWHITE	**
## VICTIM_RACEWHITE HISPANIC	***
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEASIAN / PACIFIC ISLANDER	***
## PERP_RACEBLACK:VICTIM_RACEASIAN / PACIFIC ISLANDER	
## PERP_RACEBLACK HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER	
## PERP_RACEWHITE:VICTIM_RACEASIAN / PACIFIC ISLANDER	***
## PERP_RACEWHITE HISPANIC:VICTIM_RACEASIAN / PACIFIC ISLANDER	
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK	***
## PERP_RACEBLACK:VICTIM_RACEBLACK	.
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK	***
## PERP_RACEWHITE:VICTIM_RACEBLACK	
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK	
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEBLACK HISPANIC	
## PERP_RACEBLACK:VICTIM_RACEBLACK HISPANIC	
## PERP_RACEBLACK HISPANIC:VICTIM_RACEBLACK HISPANIC	***
## PERP_RACEWHITE:VICTIM_RACEBLACK HISPANIC	
## PERP_RACEWHITE HISPANIC:VICTIM_RACEBLACK HISPANIC	
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE	***
## PERP_RACEBLACK:VICTIM_RACEWHITE	
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE	
## PERP_RACEWHITE:VICTIM_RACEWHITE	***
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE	
## PERP_RACEASIAN / PACIFIC ISLANDER:VICTIM_RACEWHITE HISPANIC	

```

## PERP_RACEBLACK:VICTIM_RACEWHITE HISPANIC
## PERP_RACEBLACK HISPANIC:VICTIM_RACEWHITE HISPANIC
## PERP_RACEWHITE:VICTIM_RACEWHITE HISPANIC
## PERP_RACEWHITE HISPANIC:VICTIM_RACEWHITE HISPANIC
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 59173.8  on 125  degrees of freedom
## Residual deviance:  2368.5  on  94  degrees of freedom
## AIC: 3042.8
##
## Number of Fisher Scoring iterations: 5

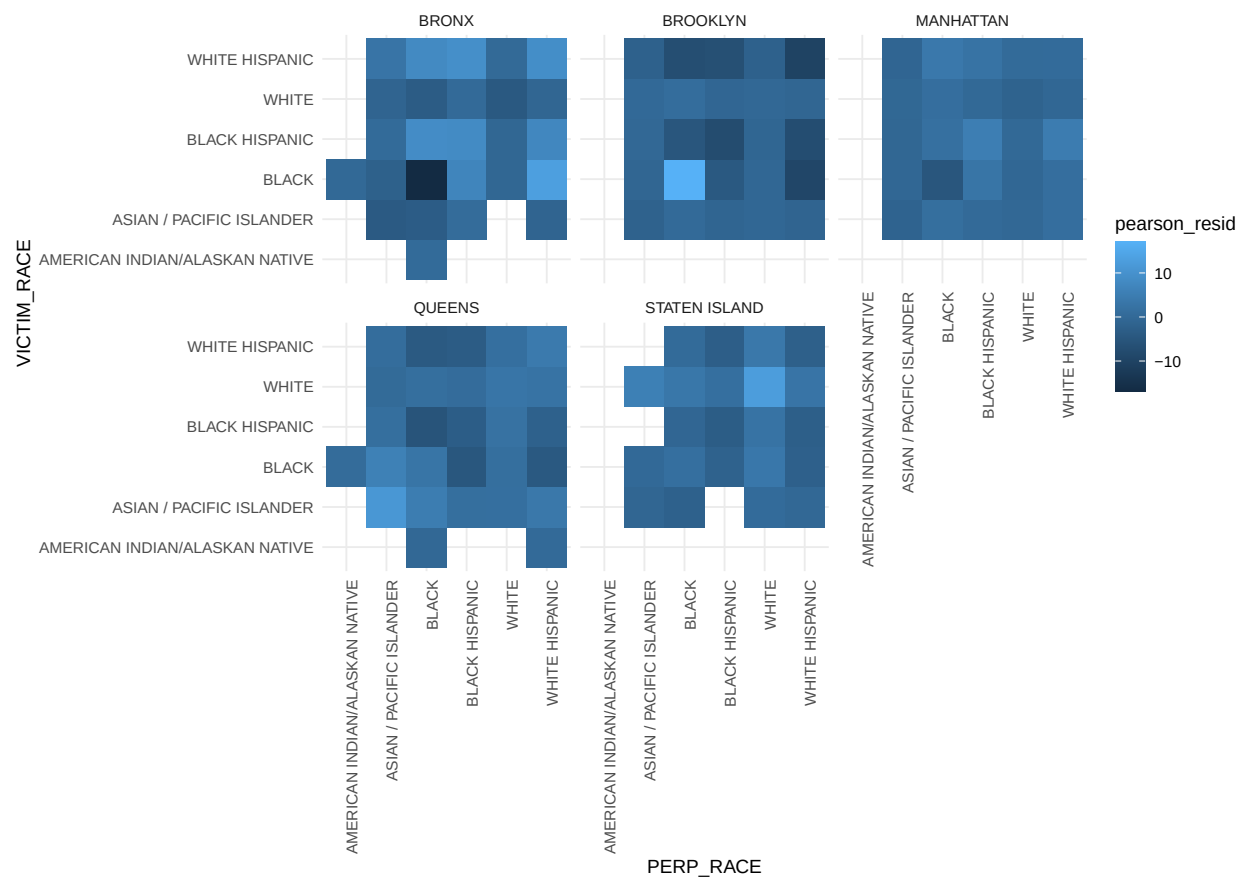
```

```

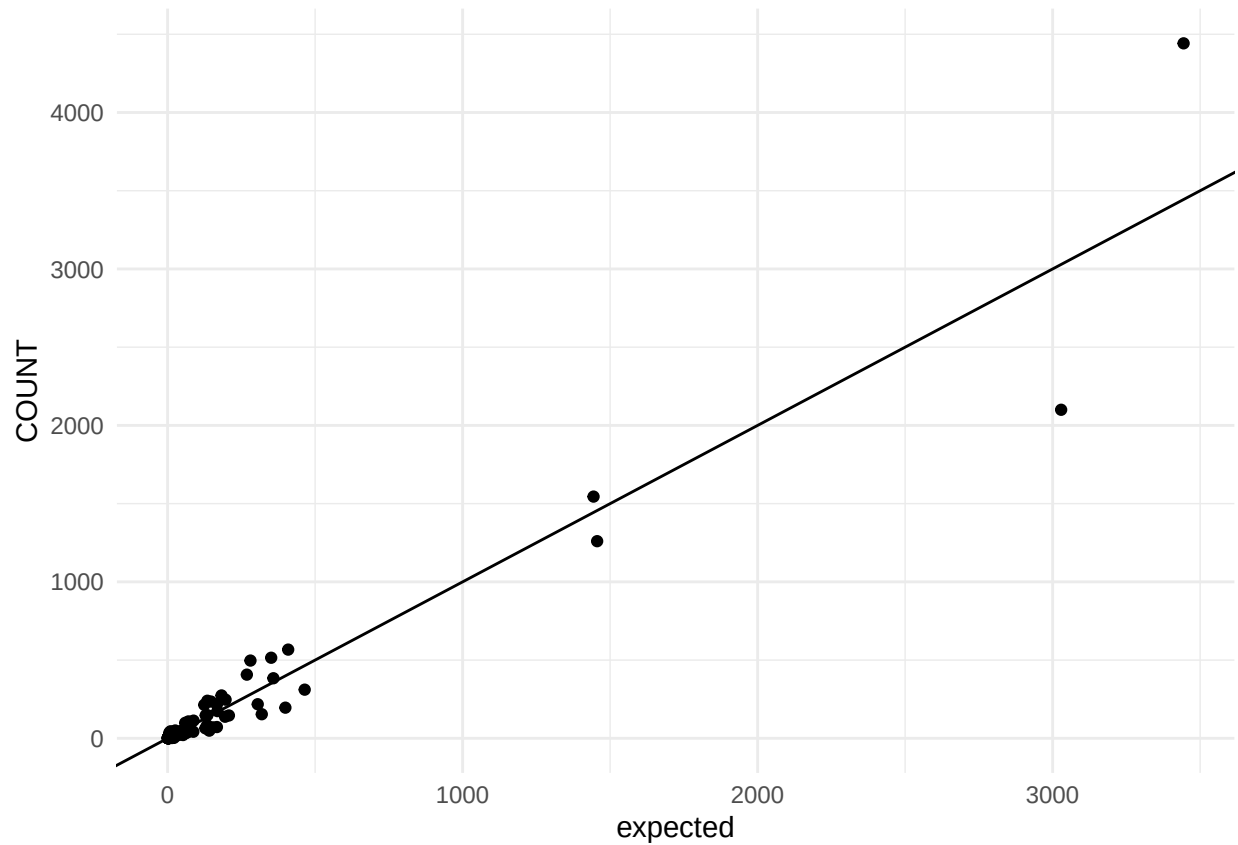
df$expected <- predict(model, type = "response")
df$pearson_resid <- residuals(model, type = "pearson")

ggplot(df, aes(PERP_RACE, VICTIM_RACE, fill = pearson_resid)) +
  geom_tile() +
  facet_wrap(~ BORO) +
  theme_minimal() +
  coord_equal() +
  theme(axis.text.x.bottom = element_text(angle=90, hjust=1))

```



```
ggplot(df, aes(expected, COUNT)) +
  geom_point() +
  geom_abline(slope = 1, intercept = 0) +
  theme_minimal()
```



Looks fairly 1-to-1 linear, which makes sense since we're comparing against the data we have trained again. To test this appropriately most likely would need to do cross-validation.

Brief comment on Bias

The analysis may be biased towards certain demographics which have a larger proportion of crimes and skew the definition of same-race crime. For example, certain ethnicity in New York may be less prevalent and therefore less likely to conduct same-race crimes, or more likely to conduct different-race crime. It's important to consider this when conducting analysis like this.